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PROBABILITY AND STATISTICS

UNIT I

PROBABILITY AND RANDOM VARIABLE

MOMENTS-MOMENT GENERATING FUNCTIONS AND THEIR PROPERTIES:

MOMENTS:

DISCRETE CASE:

Let X be discrete R.V. taking the values $x_1, x_2, \cdots x_n$ with probability mass function $p_1, p_2, \cdots p_n$ respectively then the rth moment about origin is

$$\mu_r' \left(\text{about the origin}\right) = \sum_{i=1}^{n} x_i^r \, p_i \qquad \cdots (1)$$

$$\mu_r' \left(\text{about any point } x = A\right) = \sum_{i=1}^{n} \left(x_i - A\right)^r p_i \qquad \cdots (2)$$

$$\mu_r' \left(\text{about mean}\right) = \sum_{i=1}^{n} \left(x_i - \text{Mean}\right)^r p_i \qquad \cdots (3)$$

CONTINUOUS CASE:

If X is a continuous R.V. with probability density function *f(x)* defined in the interval (*a*, *b*) then

$$\mu_r' = \int_a^b x^r \, f(x) dx \qquad \cdots (1)$$

$$\mu_r' \left(\text{about a point } x = A\right) = \int_a^b \left(x - A\right)^r f(x) dx \qquad \cdots (2)$$

$$\mu_r' \left(\text{about the mean}\right) = \int_a^b \left(x - \bar{x}\right)^r f(x) dx \qquad \cdots (3)$$

RESULTS:

DISCRETE CASE

$$\mu_1' = \sum_{i=1}^{n} x_i p_i = \text{Mean}\left(\bar{x}\right)$$

$$\mu_2' = \sum_{i=1}^{n} x_i^2 p_i = \text{Mean square value}$$

$$\mu_2 = \sum_{i=1}^{n} (x_i - \text{mean})^2 p_i = \text{Variance}$$

$$= \mu_2' - \left(\mu_1'\right)^2$$

$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4$$

CONTINUOUS CASE

$$\mu_1' = \int_a^b xf(x)dx = \text{Mean}\left(\bar{x}\right)$$

$$\mu_2' = \int_a^b x^2 f(x)dx = \text{Mean square value}$$

$$\mu_2 = \int_a^b (x - \text{Mean})^2 f(x)dx = \text{Variance}$$

$$= \mu_2' - \left(\mu_1'\right)^2$$

$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4$$

MOMENT GENERATING FUNCTION: (M.G.F)

An important device that can be used to calculate the higher moments is the moment generating function.

Definition:

Moment generating function of a random variable X about the origin is defined as

$$M_X(t) = E\left[e^{tX}\right] = \sum_X e^{tX} p(x), \text{ if } X \text{ is discrete}$$

$$= \int_{-\infty}^{\infty} e^{tX} f_X(x)dx, \text{ if } X \text{ is continuous}$$

NOTE:

1. $M_X(t) = E\left[e^{tX}\right] = 1 + t\mu_1' + \frac{t^2}{2!}\mu_2' + \frac{t^3}{3!}\mu_3' + \cdots + \frac{t^r}{r!}\mu_r' + \cdots$

   Where $\mu_r' = r^{th}$ moment about the origin $= E\left[X^r\right] =$ coefficient of $\frac{t^r}{r!}$ in $M_X(t)$.

2. $\mu_r' = \left\{\frac{d^r}{dt^r}\left[M_X(t)\right]\right\}_{t=0}$

(a) PROPERTIES OF M.G.F:

1. Let $Y = aX + b$, where X is a R.V with moment generating function $M_X(t)$. Then

   $$M_Y(t) = E\left[e^{tY}\right] = E\left[e^{t(aX+b)}\right] = E\left[e^{taX} e^{bt}\right]$$
   $$= e^{bt} E\left[e^{Xat}\right] = e^{bt} M_X(at)$$

2. $M_{cX}(t) = E\left[e^{cXt}\right] = E\left[e^{X(ct)}\right] = M_X(ct)$, where c is a constant.

3. If X and Y are two independent random variables, then $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$

   Proof:

$$M_{X+Y}(t) = E\left[e^{t(X+Y)}\right] = E\left[e^{tX+tY}\right] = E\left[e^{tX}e^{tY}\right]$$
$$= E\left[e^{tX}\right]E\left[e^{tY}\right] \qquad [\because \text{X and Y are independent}]$$
$$= M_X(t)\cdot M_Y(t)$$

PROBLEMS BASED ON M.G.F:

1. Find the moment generating function of the RV X whose probability function $P(X=x) = \frac{1}{2^x}$, $x = 1,2,\cdots$. Hence find its mean.

Solution:

Given: $P(X=x) = \frac{1}{2^x}$, $x = 1,2,\cdots$

To find: M.G.F

$$M_X(t) = E\left(e^{tX}\right)$$
$$= \sum_{x=1}^{\infty} e^{tX}\, p(x)$$
$$= \sum_{x=1}^{\infty} e^{tX}\,\frac{1}{2^x} = \sum_{x=1}^{\infty}\left(\frac{e^t}{2}\right)^x$$
$$= \frac{e^t}{2} + \left(\frac{e^t}{2}\right)^2 + \left(\frac{e^t}{2}\right)^3 + \cdots$$
$$= \frac{e^t}{2}\left[1 + \frac{e^t}{2} + \left(\frac{e^t}{2}\right)^2 + \cdots\right]$$
$$= \frac{e^t}{2}\left[1 - \frac{e^t}{2}\right]^{-1} = \frac{e^t}{2}\left[\frac{2-e^t}{2}\right]^{-1} = \frac{e^t}{2-e^t}$$

To find: Mean

$$\text{Mean} = E[X] = \mu_1' = \left\{\frac{d}{dt}\left[M_X(t)\right]\right\}_{t=o}$$
$$= \left\{\frac{d}{dt}\left[\frac{e^t}{2-e^t}\right]\right\}_{t=o}$$
$$= \left\{\frac{\left(2-e^t\right)e^t - e^t\left(-e^t\right)}{\left(2-e^t\right)^2}\right\}_{t=0}$$
$$= \left\{\frac{(2-1)(1)-(1)(-1)}{(2-1)^2}\right\} = 2$$

2. For the triangular distribution $f(x) = \begin{cases} x, & 0 < x \le 1 \\ 2-x, & 1 \le x < 2 \\ 0, & \text{otherwise} \end{cases}$ find the mean, variance and the moment generating function (MGF) also find cdf of F(x).

Solution:

Given: $f(x) = \begin{cases} x, & 0 < x \le 1 \\ 2-x, & 1 \le x < 2 \\ 0, & \text{otherwise} \end{cases}$

To find : Mean

$$\text{Mean} = E(X) = \int_{-\infty}^{\infty} x f(x)dx \quad \cdots(1)$$
$$= \int_0^1 (x)(x)dx + \int_1^2 (x)(2-x)dx$$
$$= \int_0^1 x^2 dx + \int_1^2 \left(2x - x^2\right)dx$$

To find: Variance

$$Var(X) = E\left(X^2\right) - [E(X)]^2 \quad \cdots(2)$$
$$\text{Now, } E\left(X^2\right) = \int_{-\infty}^{\infty} x^2 f(x)dx$$
$$= \int_0^1 x^2 x dx + \int_1^2 x^2(2-x)dx$$

$$= \left[\frac{x^3}{3}\right]_0^1 + \left[\frac{2x^2}{2} - \frac{x^3}{3}\right]_1^2$$

$$= \frac{1}{3} + \left[\left(4 - \frac{8}{3}\right) - \left(1 - \frac{1}{3}\right)\right] = 1$$

$$= \int_0^1 x^3 dx + \int_1^2 \left(2x^2 - x^3\right) dx$$

$$= \left[\frac{x^4}{4}\right]_0^1 + \left[\frac{2x^3}{3} - \frac{x^4}{4}\right]_1^2$$

$$= \left(\frac{1}{4} - 0\right) + \left(\frac{16}{3} - \frac{16}{4}\right) - \left(\frac{2}{3} - \frac{1}{4}\right) = \frac{7}{6}$$

$$\therefore (2) \Rightarrow Var(X) = E\left(X^2\right) - [E(X)]^2 = \frac{7}{6} - 1 = \frac{1}{6}$$

To find: M.G.F

$$M_X(t) = E\left(e^{tX}\right) = \int_{-\infty}^{\infty} e^{tx} f(x) dx = \int_0^1 x e^{tx} dx + \int_1^2 (2 - x) e^{tX} dx$$

$$= \left[x\frac{e^{tx}}{t} - \frac{e^{tx}}{t^2}\right]_0^1 + \left[(2 - x)\frac{e^{tx}}{t} - (-1)\frac{e^{tx}}{t^2}\right]_1^2$$

$$= \left[\left(\frac{e^t}{t} - \frac{e^t}{t^2}\right) - \left(0 - \frac{1}{t^2}\right)\right] + \left[\left(0 + \frac{e^{2t}}{t^2}\right) - \left(\frac{e^t}{t} + \frac{e^t}{t^2}\right)\right]$$

$$= \frac{e^{2t}}{t^2} - \frac{2e^t}{t^2} + \frac{1}{t^2} = \frac{1}{t^2}\left[e^t - 1\right]^2$$

To find: The cdf of F(x)

$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(x) dx$$

(i) If $x \le 0$, then $F(x) = 0$

(ii) If $0 < x \le 1$, then $F(x) = \int_0^x x dx = \left[\frac{x^2}{2}\right]_0^x = \frac{x^2}{2}$

(iii) If $1 \le x \le 2$, then $F(x) = \int_0^1 x dx + \int_1^x (2 - x) dx$

$$= \left[\frac{x^2}{2}\right]_0^1 + \left[2x - \frac{x^2}{2}\right]_1^x$$

$$= \frac{1}{2} + 2x - \frac{x^2}{2} - 2 + \frac{1}{2} = 2x - -\frac{x^2}{2} - 1$$

(iv) If $x > 2$, then $F(x) = \int_0^1 x dx + \int_1^2 (2 - x) dx + \int_2^x 0 dx$

$$= \left[\frac{x^2}{2}\right]_0^1 + \left[2x - \frac{x^2}{2}\right]_1^2$$

$$= \frac{1}{2} + (4 - 2) - \left(2 - \frac{1}{2}\right) = 1$$

<u>EXERCISE: 1.3</u>

1. If the MGF of a uniform distribution for a random variable X is $\frac{1}{t}\left(e^{5t} - e^{4t}\right)$, find $\mathrm{E}(\mathrm{X})$.

2. The moment generating function of a random variable X is given by $M(t)=e^{3(e^t-1)}$. What is $P[X=0]$?

3. An experiment succeeds twice as often as it fails. Find the chance that in the next 4 trials, there shall be at least one success.

4. A random variable X has cdf $F_X(x)=\begin{cases} 0 & ; \quad x<1 \\ \frac{1}{2}(x-1) & ; \quad 1\le x<3 \\ 1 & ; \quad x\ge 3 \end{cases}$. Find the pdf of X and the expected value of X.

5. A random variable X has pdf $f_X(x)=\begin{cases} kx^2e^{-x} & ; x>0 \\ 0 & \text{otherwise} \end{cases}$. Find the rth moment of X about origin. Hence find the mean and variance.

6. If the probability density of X is given by $f(x)=\begin{cases} 2(1-x) & \text{for } 0<x<1 \\ 0 & \text{otherwise} \end{cases}$, find its rth moment. Hence evaluate $E\left[(2X+1)^2\right]$.

7. Find MGF corresponding to the distribution $f(\theta)=\begin{cases} \frac{1}{2}e^{-\theta/2} & \theta>0 \\ 0 & \text{otherwise} \end{cases}$ and hence find its mean and variance.

8. Find the M.G.F of the random variable X having the probability density function $f(x)=\begin{cases} \frac{x}{4}e^{\frac{-x}{2}}, x>0 \\ 0, \quad \text{elsewhere} \end{cases}$. Also deduce that first four moments about the origin.

Discussion Questions:

1. What are moment generating functions MGFS for discrete random variables?

2. What is a central moment of a discrete variable?